COMPANY LEAVE MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

SHREYA AMUDHU N R

ROLL NO. 240701505

in partial fulfilment of the End Semester Examination for the course

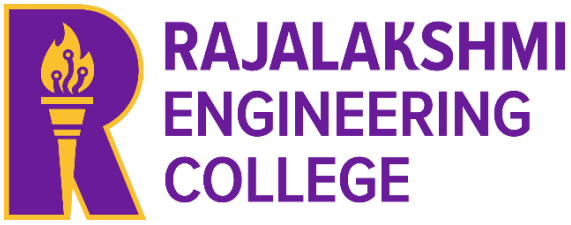
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

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RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report titled "EcoCycle – E-Waste Collection & Recycling Management" is the bonafide work of Jenell S G (Roll No. 240701212), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

SIGNATURE	**SIGNATURE**
Dr. J. MANORANJANI	Dr. J. Jinu Sophia
HEAD OF THE DEPARTMENT	Associate Professor
Department of CSE	Department of CSE
Rajalakshmi Engineering College	Rajalakshmi Engineering College
Thandalam – 602 105	Thandalam – 602 105

Submitted for the End Semester Examination (Individual Project) held on _____

INTERNAL EXAMINER	**EXTERNAL EXAMINER**

DECLARATION

I, Shreya Amudhu N.R. (Roll No. 2116240701505), hereby declare that the project report titled "Company Leave Management System (CLMS)", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, feature, code snippet or external resource used in this work has been duly acknowledged in the report

Place: Chennai
Date:

SIGNATURE OF THE STUDENT
SHREYA AMUDHU N R
ROLL NO. 240701505

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SHREYA AMUDHU N R

ABSTRACT

Administrative tracking of employee leaves in small and medium-sized organisations is frequently plagued by fragmented channels like verbal requests, text threads, and standalone spreadsheets. Without a single record of absolute balances, request flows, and automated checking, operations face calculation errors, duplicate balance assignments, delayed manager responses, and a complete lack of security.

Company Leave Management System (CLMS) is a centralised human resources application built on Zoho Creator. Its structural foundations leverage seven baseline forms: Department Details, Employee Details, Leave Type Details, Leave Balance, Leave Request, Extra Leave Request, and Holiday Details. A Deluge form workflow enforces transaction integrity by verifying balance tracking records dynamically, checking that total requested days do not exceed remaining allocations, and blocking duplicate profile entries. New requests initialize automatically into a Submitted state, validate timelines, trigger instantaneous email notifications to reporting managers, and invoke a stage-by-stage Blueprint workflow (Submitted, Under Review,Approved,Rejected).

Operational tracking is delivered through ten custom reports (seven lists, two kanban layouts, one multi-axis calendar view) and a master Dashboard interface aggregating critical organizational metrics. Seamless platform integrations dynamically search and populate employee profiles as Leads into Zoho CRM and trigger automatic mail summaries via Zoho Mail. The system operates cleanly via local platform security configurations, managing operational separation across four primary organizational tiers: CEO, HR Admin, Manager, and Employee.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Leave Management; Workflow Automation; Role-Based Access Control.

PROJECT AT A GLANCE

Attribute	Details
Project Title	Company Leave Management System (CLMS)
Application Domain	Human Resources – employee leave collection and tracking management
Student Name & Register No.	Shreya Amudhu N.R. – 2116240701505
Zoho Creator Edition	Trial Edition
Application Name (Link Name)	company-leave-management-system
Published / Live URL	https://creatorapp.zoho.in/shreya.amudhu.2024.cse_rajalakshmi/clms
Demo Credentials (per role)	Owner account only
Number of Forms / Subforms	7 forms (Department, Employee, Leave Type, Balance, Request, Extra Leave, Holiday); 0 subforms
Number of Reports (by type)	10 reports (7 List views, 2 Kanban views, 1 Calendar view) and 3 dashboard charts
Number of Pages / Dashboards	1 dashboard page (CLMS Page Dashboard)
Number of Workflows	Form: 7 | Schedule: 1 | Approval: 1 | Blueprint: 1 | Batch: 0
Deluge Functions / Actions	12 distinct form-workflow and automation scripts
Integrations	Zoho CRM (Contact creation and sync); Zoho Mail (sendmail engine)
Roles & Profiles Defined	4 primary roles defined (CEO, HR Admin, Manager, Employee)
Environments Used	Single application instance (development/sandbox environment)
Date of Production Deployment	Not deployed to an external production environment (trial sandbox verified)

TABLE OF CONTENTS

ABSTRACT iv
PROJECT AT A GLANCE v
LIST OF TABLES ix
LIST OF FIGURES x
LIST OF ABBREVIATIONS xi
INTRODUCTION 1
1.1 Background 1
1.2 Problem Statement 1
1.3 Objectives 2
1.4 Scope of the Project 2
1.5 Justification for a Low-Code Approach 2
LITERATURE SURVEY AND SYSTEM STUDY 3
2.1 Study of the Existing System 3
2.2 Limitations of the Evidence Available 3
2.3 Low-Code versus Traditional Development 4
2.4 Literature Survey 4
2.5 Emerging Trends in Low-Code Development 4
2.6 Proposed System 5
REQUIREMENT ANALYSIS 6
3.1 Stakeholders and User Roles 6
3.2 Functional Requirements 6
3.3 Non-Functional Requirements 7
3.4 Hardware and Software Requirements 7
3.5 Use Case Model 8
SYSTEM DESIGN 10
4.1 System Architecture 10
4.2 Data Model 10
4.3 Form Design 11
4.4 Process and Workflow Design 13
4.5 User Interface and Navigation Design 14
4.6 Access Control Design 14
IMPLEMENTATION 15
5.1 Application Setup 15
5.2 Forms and Data Capture 15
5.3 Reports 19

5.4 Pages and Dashboards 20
5.5 Deluge Scripting 21
5.6 Workflows and Automation 23
5.7 Integrations 24
5.8 User Management and Role-Based Access 24
5.9 AI and Analytics Features 24
5.10 Pro-Code Extension 25
5.11 Mobile and Portal Access 25
TESTING AND DEBUGGING 26
6.1 Testing Strategy 26
6.2 Test Cases 26
6.3 Debugging Deluge Scripts 27
6.4 Defect Log 27
6.5 User Acceptance Testing 28
DEPLOYMENT, SECURITY AND GOVERNANCE 29
7.1 Environments and Deployment Strategy 29
7.2 Performance Optimisation 29
7.3 Security Features 29
7.4 Governance and Administration 30
RESULTS AND DISCUSSION 31
8.1 Application Output 31
8.2 Achievement of Objectives 31
8.3 Comparison with Existing System 32
8.4 Limitations of the Low-Code Approach 32
8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum 32
CONCLUSION AND FUTURE ENHANCEMENTS 33
9.1 Conclusion 33
9.2 Future Enhancements 33
9.3 Learning Outcomes and Career Relevance 33
REFERENCES 34
APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST 35
APPENDIX B – COURSE OUTCOME MAPPING 36
APPENDIX C – COMPLETE DELUGE SOURCE CODE 37
APPENDIX D – USER MANUAL 40
D.1 Accessing the Application 40
D.2 Instructions for HR Admin 40
D.3 Instruction for Employees 40

D.4 Instruction for Managers 40
APPENDIX E – EVALUATION RUBRIC..... 41
APPENDIX F – VIVA-VOCE RECORD SHEET..... 41

LIST OF TABLES

Table 2.1: Low-Code and Traditional Development Comparison 4
Table 2.2: Literature Survey..... 4
Table 3.1: Stakeholders..... 6
Table 3.2: Functional Requirements 6
Table 3.3: Non-Functional Requirements 7
Table 3.4: Hardware and Software Requirements 7
Table 3.5: Use Case 8
Table 4.1: Entities and Relationships..... 11
Table 4.2-4.8: Form Field Specifications 11
Table 4.9: Role-Permission Matrix 14
Table 5.1: Reports..... 19
Table 5.2: Deluge Scripts..... 21
Table 5.3: Workflows 23
Table 5.4: Integrations 24
Table 6.1: Test Cases 26
Table 6.2: Defect Log 27
Table 7.1: Deployment 29
Table 7.2: Security.....30
Table 8.1: Result.....31
Table 8.2: Existing System versus Proposed Application 32
Table A.1: Feature Coverage..... 35
Table B.1: Mapping of Course Outcomes..... 36

LIST OF FIGURES

Figure 2.1: Existing Manual Process 3
Figure 3.1: Use Case Diagram 8
Figure 4.1: System Architecture 10
Figure 4.2: ER Diagram 11
Figure 4.3: Leave Approval Workflow 13
Figure 4.4: Dashboard Navigation 14
Figure 5.1: Department Details Form 15
Figure 5.2: Employee Details Form 16
Figure 5.3: Leave Type Details Form 16
Figure 5.4: Leave Balance Form 17
Figure 5.5: Leave Request Form 17
Figure 5.6: Extra Leave Request Form 18
Figure 5.7: Holiday Details Form 18
Figure 5.8: Dashboard 20
Figure 5.9: Roles Hierarchy 24

LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRM	Customer Relationship Management
DS	Deluge Script
ER	Entity–Relationship
LCAP	Low-Code Application Platform
IST	Indian Standard Time
OAuth	Open Authorization
UAT	User Acceptance Testing
UI	User Interface
URL	Uniform Resource Locator
WF	Workflow
Zia	Zoho's AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Human-resources departments in small and medium-sized organizations manage employee information, leave policies, annual leave allowances, holidays and leave applications. The process involves employees who submit requests, reporting managers who review them, HR personnel who maintain employee records and balances, and management users who need a current view of employee availability.

In a manual arrangement, an employee may request leave verbally, through WhatsApp, by phone or by email. A manager may approve informally or forward the request to HR. HR records the leave in a spreadsheet or paper register and calculates used and remaining days. At year end, HR consolidates records for administration and payroll-related work. When information is stored in separate files and channels, it is difficult to ensure that every participant uses the latest information.

The project models a fictional mid-sized company using the email domain company.com. Sample configuration contains eight departments (seven active and one inactive), ten employees, eight leave types, ten leave balances, ten leave requests, six extra-leave requests and eight holidays. The application is developed using Zoho Creator and Deluge..

1.2 Problem Statement

- The project addresses the lack of a centralised, controlled and automated process for employee leave administration. Manual records can become inconsistent, balances may be calculated incorrectly, approval actions may be delayed, and employees may not be able to track request status. Shared spreadsheets also provide limited access control and management visibility.
- Employee, leave-type and balance information may be duplicated across spreadsheets.
- Manual calculations can produce incorrect used and remaining balances.
- The process does not reliably prevent requests exceeding available leave.
- Managers may overlook pending requests because reminders are not centralised.
- Shared files may expose personal information and provide no live dashboard or clear audit trail.

1.3 Objectives

1. Design a normalised data model and forms for departments, employees, leave types, balances, holidays and leave requests.
2. Automate validation, balance updates, notifications and reminders using Deluge and workflows.
3. Implement leave approval through a Blueprint and extra-leave approval through a separate process.
4. Provide role-based reports, calendar and kanban views, and a dashboard.
5. Integrate employee records with Zoho CRM and use email notifications.
6. Secure the application through roles, permission profiles and field-level restrictions.

1.4 Scope of the Project

In scope: three data forms (E-Waste Entry, Collection Request and Recycling Tracking); form workflows for priority calculation, status initialisation, date validation and e-mail notification; four reports; the EcoCycle Dashboard page; a connection to Zoho CRM that creates a lead for each collection request; a Sentiment field on the Collection Request form; a Custom API, EcoCycle Summary API; and the EcoCycle Data Assistant chat agent with read-only access.

Out of scope or not yet built: the Zoho Creator REST API integration (planned as a future enhancement and not implemented), Google Maps integration, automated collection scheduling, SMS notifications, QR-code identification and real-time collection tracking. These are listed in Chapter 9. Roles and permission profiles were not described.

1.5 Justification for a Low-Code Approach

Zoho Creator was selected because the application requires forms, lookup relationships, reports, a dashboard, approval routing, notifications and role ermissions. These capabilities are available as built-in components, reducing the need to separately develop a database, backend, frontend, authentication, hosting and mobile application. Deluge is used for business-specific validation, balance updates, email actions and CRM synchronisation. Built-in hosting, mobile access, permission profiles, field-level controls and OAuth-based CRM integration support the requirements. HR staff can maintain many fields and reports without modifying a traditional codebase [1].

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The employee communicates a leave request to a manager through verbal discussion, WhatsApp, telephone or email. The manager gives an informal response or forwards the request to HR. HR enters it in an Excel workbook or paper register, updates leave usage manually and calculates remaining allowance. At year end, HR consolidates the information.

1. Employee sends a request through an informal channel.
2. Manager reviews and communicates a decision or forwards it to HR.
3. HR enters the request in a spreadsheet or paper register.
4. HR manually calculates used and remaining leave days.
5. HR consolidates records for year-end administration.



Figure 2.1: Proposed Flow of the Existing System

2.2 Limitations of the Evidence Available

- No single source of truth for employee, leave type and balance information.
- Manual calculations may create errors and disputes.
- No consistent automatic prevention of over-booking or duplicate yearly balances.
- Delayed approvals due to lack of reminders and structured tracking.
- Limited employee visibility into request status.
- Shared files may expose personal data to unauthorised users.
- No live dashboard for management.
- Holiday and leave-policy information may be scattered.
- Extra-leave requests lack a consistent approval process.
- No automatic CRM/contact synchronisation.

2.3 Low-Code versus Traditional Development

Table 2.1: Low-Code and Traditional Development Comparison

Criterion	Traditional Development	Zoho Creator Low-Code
Development effort	Separate database, backend, frontend, login and hosting implementation.	Built-in forms, reports, dashboards, workflows and permissions.
Skills	Programming, database, web and infrastructure skills.	Creator configuration and Deluge knowledge.
Hosting	Hosting, servers, backups and deployment must be arranged.	Hosting handled by platform.
Mobile	May require separate mobile implementation.	Creator mobile app available.
Security	Authentication and access rules must be coded.	Permission profiles, roles and field-level controls built in.
Maintenance	Code changes and deployment may be needed.	HR administrators can maintain many fields and reports.
Customisation	Greater architectural freedom.	Bounded by platform features and scripting.
Cost	Development and infrastructure costs.	Creator subscription or education programme; exact cost depends on edition.

2.4 Literature Survey

The project description suggests the following five resources. Bibliographic details not provided in the source description are marked for verification rather than invented.

Table 2.2: Summary of Literature Survey

No.	Source	Title / Resource	Relevance
1	Zoho Corporation	Zoho Creator Help Documentation [1]	Forms, reports, pages, workflows and platform configuration.
2	Zoho Corporation	Deluge Script Reference Guide [2]	Scripting for validation, data retrieval, loops, email and CRM.
3	Zoho Corporation	Blueprint and Approval Process Help [3]	Stage-based leave review and extra-leave approval.
4	Zoho Corporation	Zoho CRM API / Creator–CRM Integration Documentation [4]	Create or update employee contacts.
5	W. Rosam and O. Kalnichevski	Low-Code Development: A Practical Guide, BPB Publications [5]	Practical low-code development background.

2.5 Emerging Trends in Low-Code Development

Low-code platforms support citizen development, workflow automation, reusable components, cloud hosting and integration between business applications. CLMS applies these trends through visual forms and reports, Deluge validation, scheduled reminders, approval routing and CRM synchronisation. AI-assisted building and predictive analytics are emerging capabilities, but the supplied export does not show Zia or other AI features. The dashboard uses conventional KPIs, charts, calendar and kanban reports.

2.6 Proposed System

CLMS centralises leave information in Zoho Creator. HR maintains master records and yearly balances. Employees submit requests through a form. Deluge validates that a matching balance exists and that requested days do not exceed the remaining allowance. The manager receives an email and reviews the request through a Blueprint. Approval updates used days and notifies the employee; rejection updates status and sends an email. A daily schedule reminds managers about pending requests. Extra leave uses a separate approval process. Reports and a dashboard provide visibility, while role permissions limit access.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Responsibilities
CEO / Administrator	Top-level role.	Administrative oversight according to assigned profile.
HR Admin	HR administrator.	Maintain departments, employees, leave types, holidays, balances and leave administration.
Manager	Reporting manager.	Review requests, add comments, approve/reject and view relevant reports.
Employee	Staff member.	Submit leave and extra-leave requests and track permitted status information.

3.2 Functional Requirements

Table 3.2: Functional Requirements

ID	Requirement	Role	Priority
FR-01	Maintain departments, employees, manager hierarchy, leave types and holidays.	HR Admin	High
FR-02	Auto-fill manager email when a manager is selected.	HR Admin	Medium
FR-03	Create yearly balances; fill allocated days; calculate used and remaining days.	HR Admin	High
FR-04	Prevent duplicate employee/type/year balance records.	HR Admin	High
FR-05	Allow employees to submit leave requests with dates, type and reason.	Employee	High
FR-06	Block requests when no balance exists or days exceed remaining balance.	Employee	High
FR-07	Email the reporting manager when a request is submitted.	System	High
FR-08	Route requests through Submitted, Under Review, Approved or Rejected stages.	Manager / HR	High
FR-09	Allow manager comments and email employee on decision.	Manager	High
FR-10	Update used days when a request is approved.	System	High
FR-11	Send daily reminders for pending requests.	System	Medium
FR-12	Support extra-leave requests and increase	Employee / Approver	Medium

ID	Requirement	Role	Priority
	allocation after approval.		
FR-13	Provide dashboard with KPIs, charts, calendar and kanban.	Manager / HR	Medium
FR-14	Synchronise employee details with Zoho CRM contacts.	System	Medium
FR-15	Restrict data and actions by role and profile.	All	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Reports and validations should respond appropriately for expected HR data; measured targets To be confirmed before submission.
NFR-02	Security	Users access only forms, reports, fields and actions permitted by role.
NFR-03	Usability	Clear forms and web access; Creator mobile app is available.
NFR-04	Availability	Cloud-hosted by Zoho Creator; service details depend on edition.
NFR-05	Maintainability	HR administrators can maintain many fields and reports through low-code configuration.
NFR-06	Auditability	Approval comments and statuses provide process information; verify platform audit settings.
NFR-07	Data Integrity	Validations prevent missing balances, excess requests and duplicate balance records.
NFR-08	Scalability	Application should accommodate growth; load-test results To be confirmed before submission.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator [TO FILL: edition]
Scripting	Deluge
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox
Integrated services	Zoho CRM and Zoho Mail
Mobile	Zoho Creator mobile app; actual mobile testing To be confirmed before submission

Category	Specification
Client hardware	Intel Core i3/i5 or above, 8 GB RAM and internet connection
API testing	Postman not reported as used; mention only if separately tested.

3.5 Use Case Model

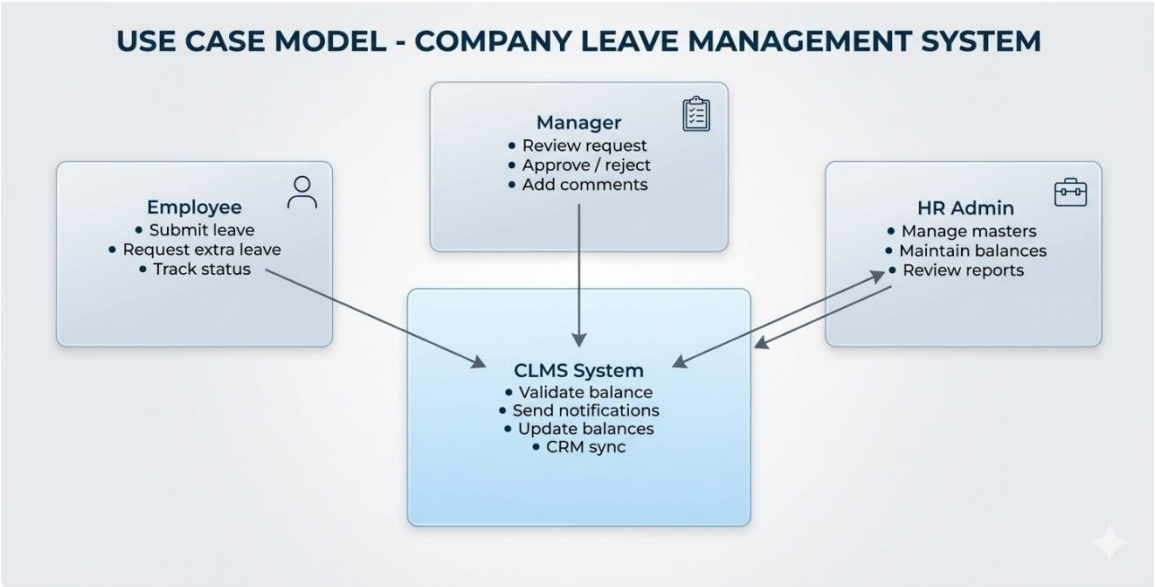


Figure 3.1: Use Case Diagram of CLMS

Table 3.5(a): Use Case Description

Use Case Item	UC-01 – Submit Leave Request
Actor	Employee
Pre-condition	Employee and matching yearly leave balance exist.
Main flow	Open Leave Request → choose employee/type → enter dates, days and reason → submit → validation → save as Submitted → email manager.
Alternate flow	No balance or insufficient balance: alert shown and submission cancelled.
Post-condition	Valid request stored and manager notification initiated.
Related FR	FR-05, FR-06, FR-07

Table 3.5(b): Use Case Description

Use Case Item	UC-02 – Review and Approve / Reject
Actor	Manager / HR Admin
Pre-condition	Request is Submitted.
Main flow	Open request → Start Review → Under Review → enter comments → Approve or Reject → update status, balance if approved and email employee.
Alternate flow	Employee is not permitted to perform Blueprint transitions.
Post-condition	Request is Approved or Rejected and employee notified.
Related FR	FR-08, FR-09, FR-10

Table 3.5(c): Use Case Description

Use Case Item	UC-03 – Manage Yearly Leave Balance
Actor	HR Admin
Pre-condition	Employee and leave type records exist.
Main flow	Select employee, type and year → allocated days auto-fill → used days calculated → remaining formula → duplicate validation → save.
Alternate flow	Duplicate employee/type/year triggers alert and cancels submission.
Post-condition	Unique yearly balance stored.
Related FR	FR-03, FR-04

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The architecture contains user, application, data and integration layers. Employees, managers, HR administrators and CEO access Zoho Creator through the web interface; the mobile app is also available. The application layer contains seven forms, ten reports, one dashboard, Deluge scripts, form workflows, a schedule, an approval process and a Blueprint. The data layer stores departments, employees, leave types, balances, requests and holidays. The integration layer connects to Zoho CRM and sends mail through Creator.

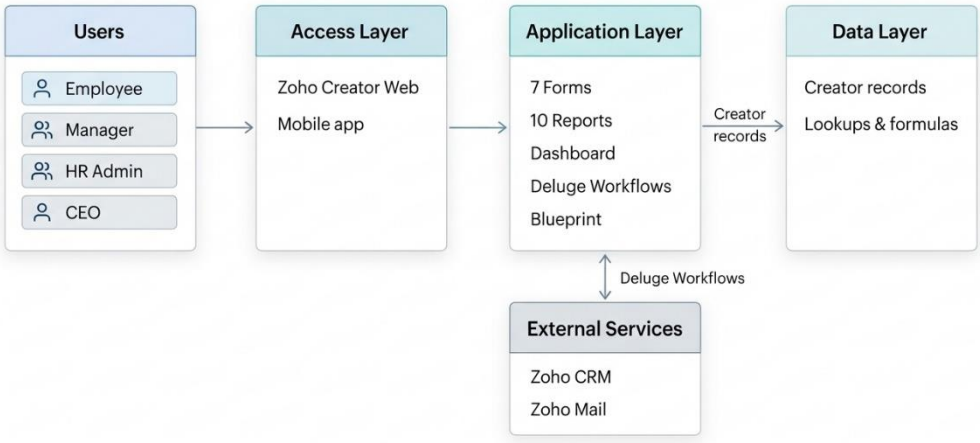


Figure 4.1: System Architecture of CLMS

4.2 Data Model

Seven forms implement the data model. Department_Details relates one-to-many with Employee_Details. Employee_Details has a self-referencing Manager lookup. Employee records relate to leave balances, leave requests and extra-leave requests. Leave_Type_Details relates to those same three transaction forms. Holiday_Details is standalone. Leave_Balance is logically matched to Leave_Request using employee, leave type and year derived from Start Date; this matching is performed in Deluge.

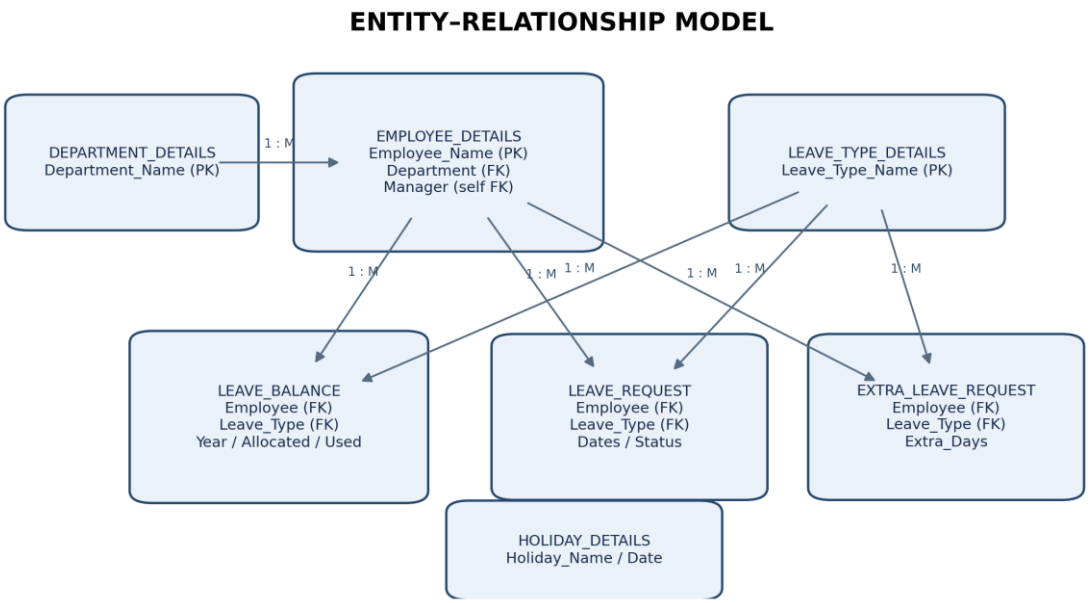


Figure 4.2: Data Model of EcoCycle

Table 4.1: Entities and Relationships

Entity	Key Fields	Relationships / Implementation
Department_Details	Department_Name, Department_Head, Department_Email, Department_Status	1:M Employee through Department lookup
Employee_Details	Employee_Name, Work_Email, Department, Manager, Manager_Email	1:M balance, leave and extra leave; self lookup for manager
Leave_Type_Details	Leave_Type_Name, Annual_Allowance, Maximum_Consecutive_Days, Paid_Leave	1:M balance, leave and extra leave
Leave_Balance	Employee, Leave_Type, Year_field, Allocated_Days, Used_Days, Remaining_Days	Lookups; formula; logical match with requests
Leave_Request	Employee, Leave_Type, Start_Date, End_Date, Total_Days, Status	Employee/type lookups; Blueprint
Extra_Leave_Request	Employee, Leave_Type, Extra_Days, Reason	Employee/type lookups; approval process
Holiday_Details	Holiday_Name, Holiday_Date, Description	Standalone reference form

4.3 Form Design

Table 4.2: Field Specification – Department_Details

Field Label	Link Name	Field Type	Mandatory	Validation / Remarks
Department Name	Department_Name	Single Line	Yes	—
Department Head	Department_Head	Single Line	No	—
Department Email	Department_Email	Single Line	No	—
Department Status	Department_Status	Drop Down	No	Active / Inactive

Table 4.3: Field Specification – Employee_Details

Field Label	Link Name	Field Type	Mandatory	Validation / Remarks
Employee Name	Employee_Name	Single Line	Yes	—
Work Email	Work_Email	Email	Yes	Max 80; personal data
Department	Department	Lookup	No	Department_Details
Designation	Designation	Single Line	No	—
Manager	Manager	Lookup	No	Self lookup to Employee_Details
Manager Email	Manager_Email	Email	No	Auto-filled; personal data

Table 4.4: Field Specification – Leave_Type_Details

Field Label	Link Name	Field Type	Mandatory	Validation / Remarks
Leave Type Name	Leave_Type_Name	Single Line	—	—
Annual Allowance	Annual_Allowance	Number	—	Days per year
Maximum Consecutive Days	Maximum_Consecutive_Days	Number	—	Stored, not enforced
Paid Leave	Paid_Leave	Checkbox / Decision Box	—	Default false

Table 4.5: Field Specification – Leave_Balance

Field Label	Link Name	Field Type	Mandatory	Validation / Remarks
Employee	Employee	Lookup	—	Employee_Details
Leave Type	Leave_Type	Lookup	—	Leave_Type_Details
Year	Year_field	Number	—	Balance year
Allocated Days	Allocated_Days	Number	—	From Annual Allowance
Used Days	Used_Days	Number	—	Default 0 / calculated
Remaining Days	Remaining_Days	Formula	—	Allocated_Days - Used_Days

Table 4.6: Field Specification – Leave_Request

Field Label	Link Name	Field Type	Mandatory	Validation / Remarks
Employee	Employee	Lookup	Yes	Employee_Details
Leave Type	Leave_Type	Lookup	Yes	Leave_Type_Details
Start Date	Start_Date	Date	Yes	All weekdays allowed
End Date	End_Date	Date	Yes	—
Total Days	Total_Days	Number	No	User-entered
Reason for Leave	Reason_for_Leave	Multi Line	Yes	—
Manager Comments	Manager_Comments	Multi Line	No	Manager decision comments
Status	Status	Drop Down	No	Default Submitted; four statuses

Table 4.7: Field Specification – Extra_Leave_Request

Field Label	Link Name	Field Type	Mandatory	Validation / Remarks
Employee	Employee	Lookup	Yes	Employee_Details
Leave Type	Leave_Type	Lookup	Yes	Leave_Type_Details
Extra Days	Extra_Days	Number	Yes	—
Reason	Reason	Multi Line	No	—

Table 4.8: Field Specification – Holiday_Details

Field Label	Link Name	Field Type	Mandatory	Validation / Remarks
Holiday Name	Holiday_Name	Single Line	Yes	—
Holiday Date	Holiday_Date	Date	Yes	—
Description	Description	Multi Line	No	—

All forms use left label placement. Add forms provide Submit; edit forms provide Update and Cancel. No subforms are used.

4.4 Process and Workflow Design

HR first creates master records and employee details. Selecting a manager fills the manager email and the employee is synchronised to CRM. HR creates yearly balances; selecting a leave type fills the allowance and calculates used days from approved requests. Duplicate validation prevents a second balance for the same employee, leave type and year.

When an employee submits a request, Deluge checks for a matching balance and compares Total Days with Remaining Days. A successful submission emails the manager. The Blueprint moves the request from Submitted to Under Review, then to Approved or Rejected. Approval updates used days and emails the employee; rejection sends a rejection email. A daily schedule reminds managers about pending requests. Extra leave follows a separate approval process..



Table 4.3: Leave Request and Approval Workflow

4.5 User Interface and Navigation Design

The supplied export carries a legacy application display name; the report title is presented as “Company Leave Management System (CLMS)”. The deployed application display name and link name should be updated to CLMS in Zoho Creator before final demonstration. Navigation contains Forms, Reports and Dashboard. Forms maintain master and transaction records. List reports provide tabular views, kanban reports group records by leave type, and the calendar displays requests between start and end dates. Dashboard includes KPIs, calendar, charts and embedded reports.

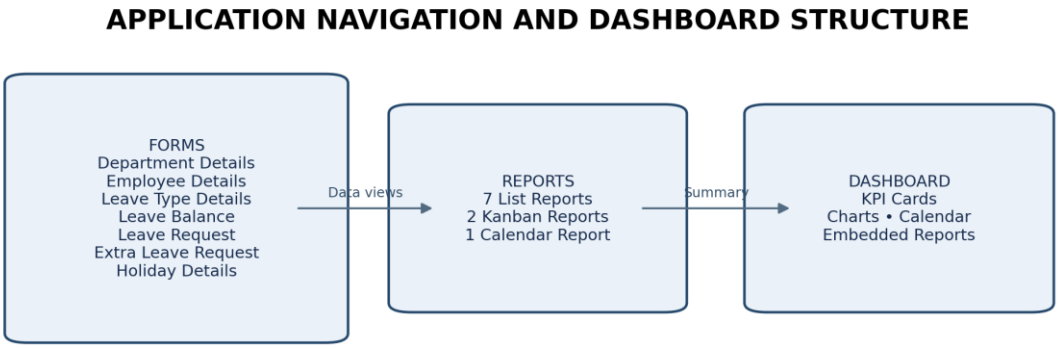


Figure 4.3: Navigation Structure of the Application

4.6 Access Control Design

Role / Profile	**Permissions / Restrictions**
CEO	Top role in hierarchy; exact access depends on assigned profile.
HR_Admin	Maintain HR master data, employee records, balances and leave administration.
Manager permissions	Start Review, Approve and Reject; view/edit request reports; selected master data view-only.
Employee permissions	Create leave and extra-leave requests; view permitted requests, calendar, holidays and leave types; no approval decisions.
Employee field restrictions	Employee Name, Work Email, Manager and Manager Email read-only; Designation and Department visible.
Administrator Permission	Dashboard tab enabled; verify live assignment.
Read / Write / Developer / Customer	Profiles present in export; verify exact assignments.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The Company Leave Management System application is built on Zoho Creator using Deluge. The export identifies version 1.0 and an export date of 31 October 2026. Settings use Asia/Kolkata time zone, dd-MMM-yyyy date format and 24-hour time. The icon displays SLM and the application display name is “Company Leave Management System”. Before submission: verify whether the app was created from scratch, copied or imported; confirm Creator edition, live URL and final application link name. The supplied export currently shows the legacy display name from the supplied export.

5.2 Forms and Data Capture

Seven forms implement master and transaction data. Department_Details stores department names, heads, emails and status. Employee_Details stores employee identity, work email, department, designation, manager and manager email. Leave_Type_Details stores allowances, maximum consecutive days and paid/unpaid status. Leave_Balance stores yearly allocated, used and remaining days. Leave_Request captures employee, leave type, dates, total days, reason, comments and status. Extra_Leave_Request captures additional days and reason. Holiday_Details stores holiday names, dates and descriptions.

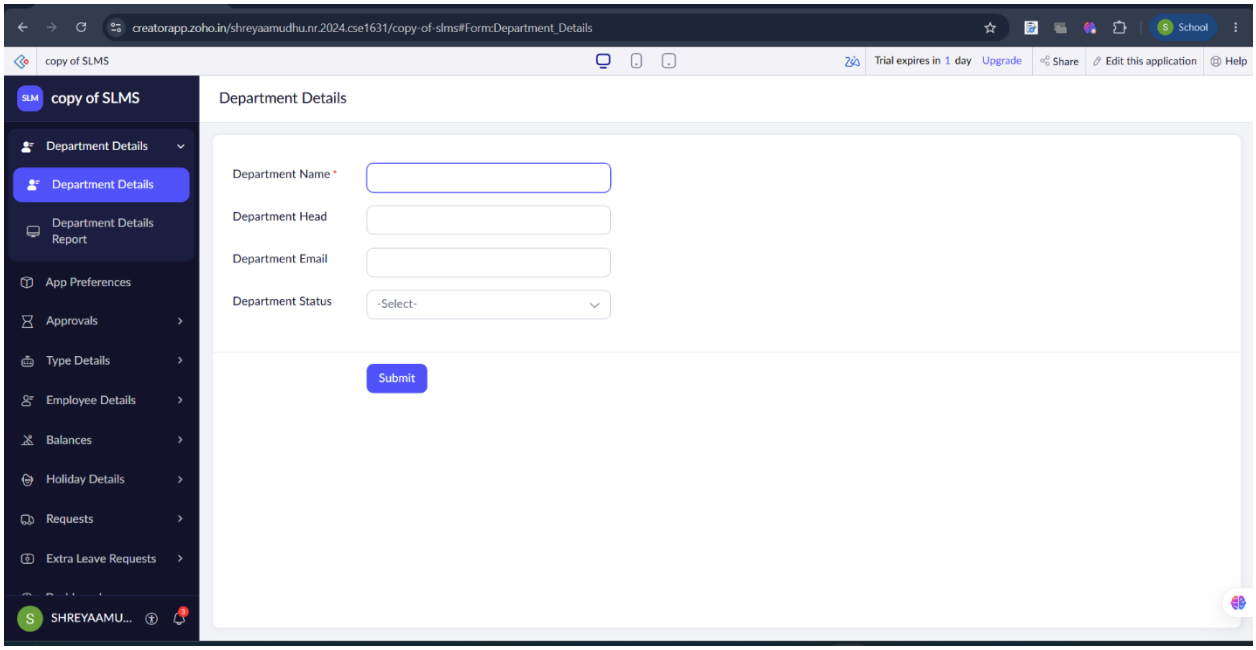


Figure 5.1: Department Details form

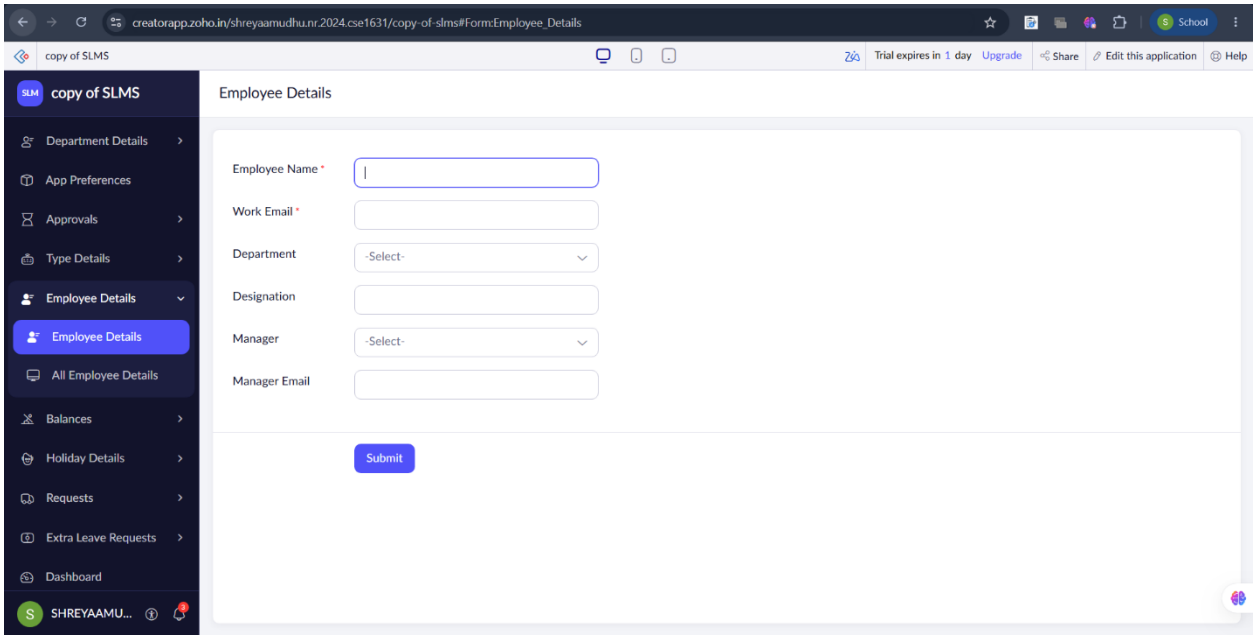


Figure 5.2: Employee Details form

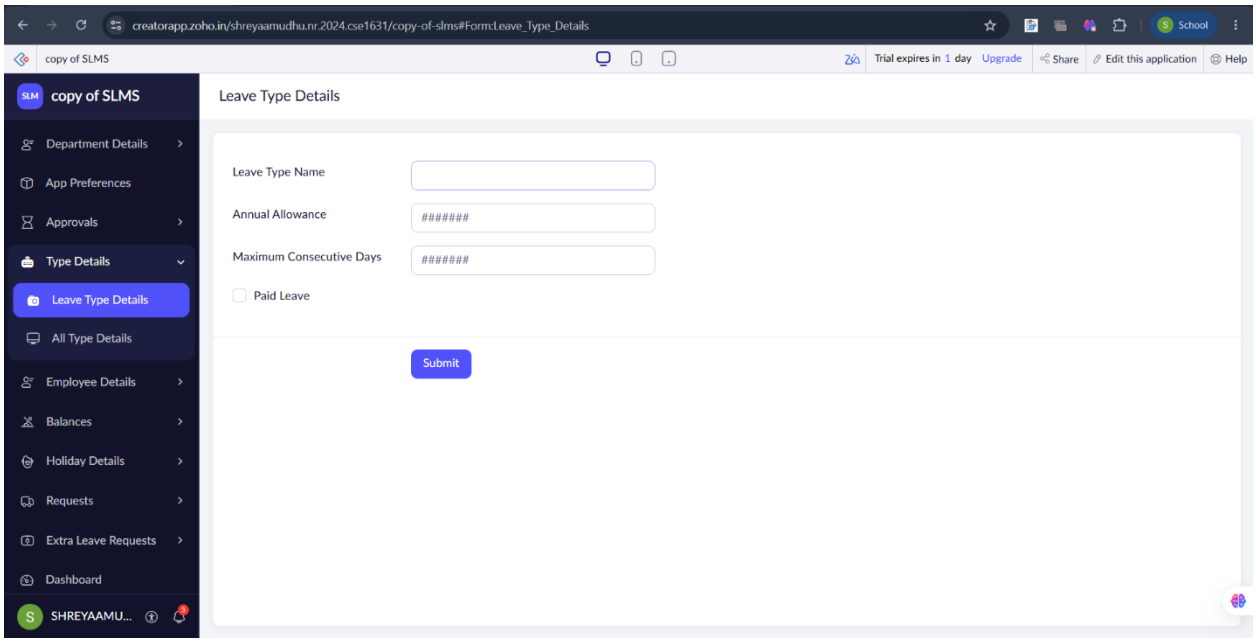


Figure 5.3: Leave Type Details form

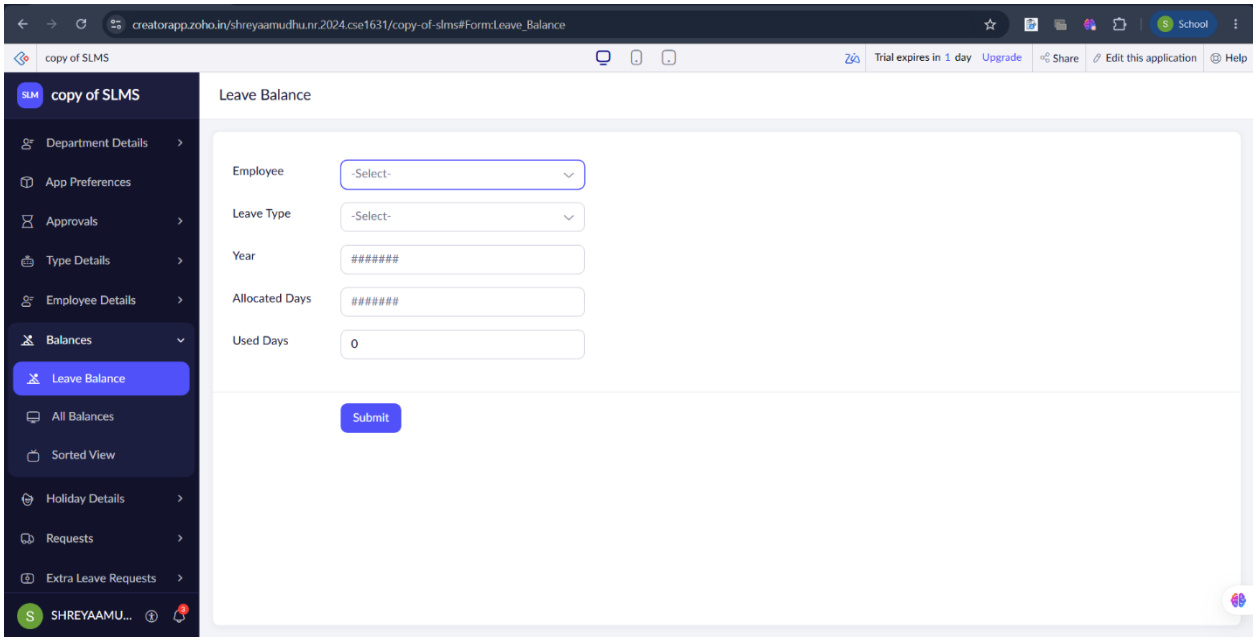


Figure 5.4: Leave Balance Details form

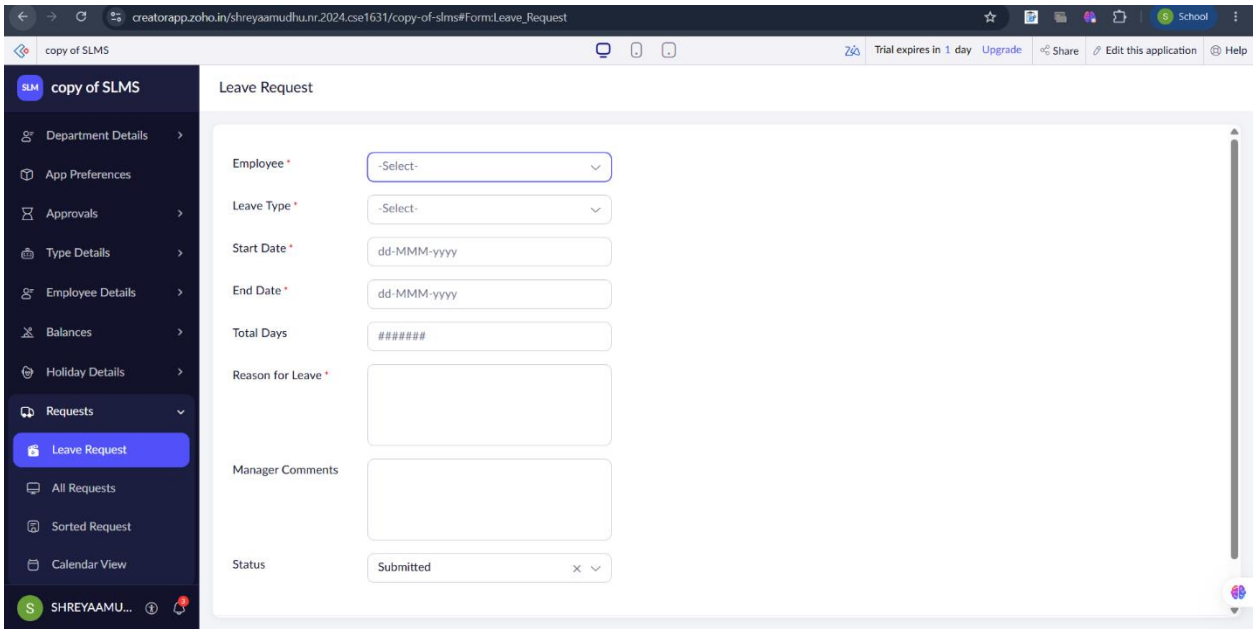


Figure 5.5: Leave Request form

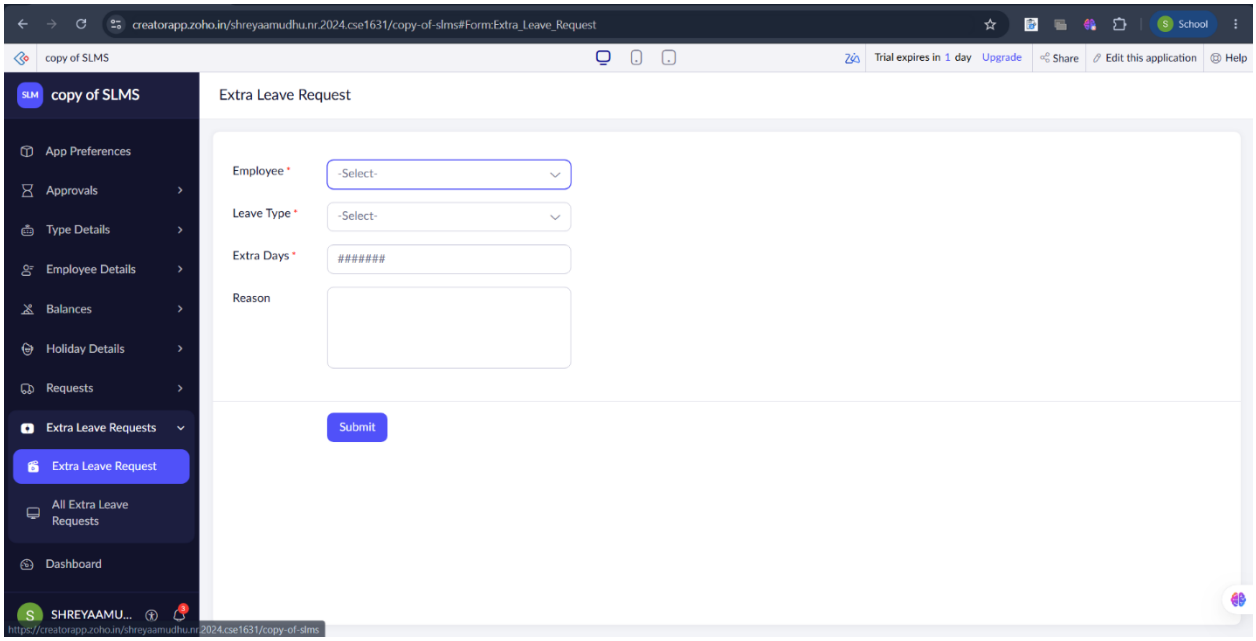


Figure 5.6: Extra Leave Request form

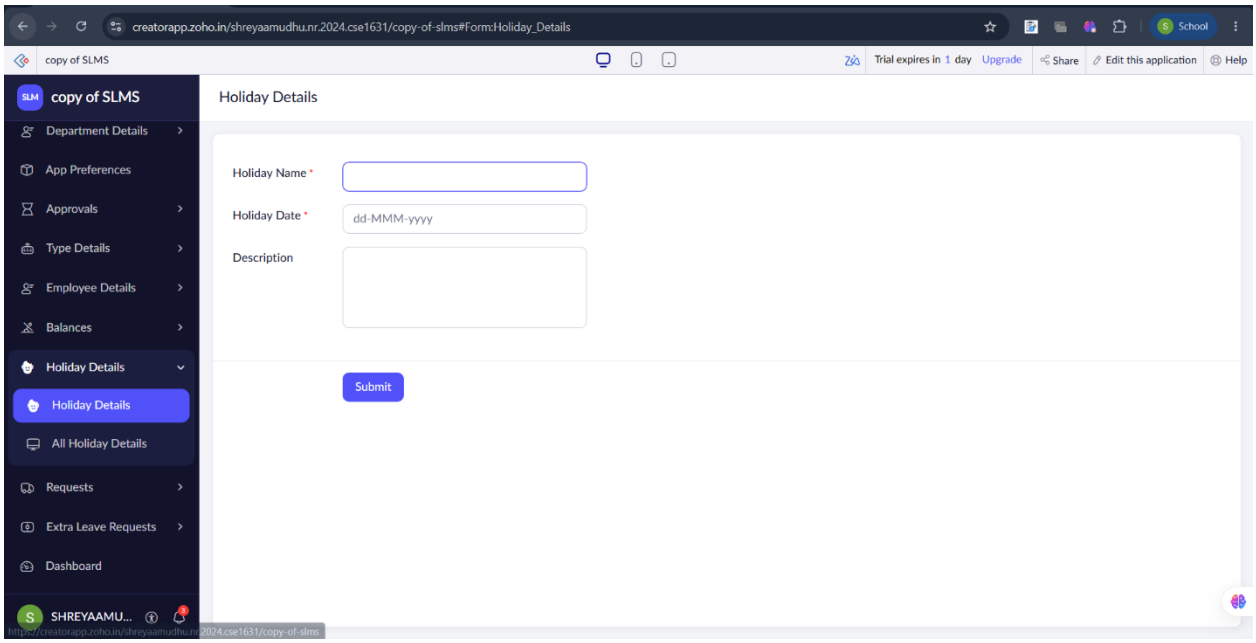


Figure 5.7: Holiday Details form

Figure 5.5: Recycling Tracking Form

5.3 Reports

List reports provide Quick View and Detail View layouts. Edit, Duplicate and Delete actions are available according to permissions.

Table 5.1: Reports Implemented

No.	Report	Type	Source / Content
1	Department Details Report	List	Department_Details: name, head, email, status
2	All Type Details	List	Leave_Type_Details: name, allowance, max consecutive days, paid leave
3	All Employee Details	List	Employee_Details: name, email, department, designation, manager
4	All Balances	List	Leave_Balance: employee, type, year, allocated, used, remaining
5	All Holiday Details	List	Holiday_Details: name, date, description
6	All Requests	List	Leave_Request: employee, type, dates, days, reason, comments, status
7	All Extra Leave Requests	List	Extra_Leave_Request: employee, type, extra days, reason
8	Sorted View	Kanban	Leave_Balance cards grouped by leave type
9	Sorted Request	Kanban	Leave_Request cards grouped by leave type
10	Calendar View	Calendar	Leave requests from Start Date to End Date, labelled by reason

5.4 Pages and Dashboards

One page named Dashboard is built using the Creator page builder. It contains three KPI panels: Total Employee, Total Department and Total Leave Request. It embeds Calendar View, a column chart plotting Start Date against End Date, and a line chart showing sum of Total Days by Status. Department Details Report and All Employee Details appear side by side, and Sorted Request appears as a full-width kanban view. The export enables the Dashboard tab only for Administrator Permission.

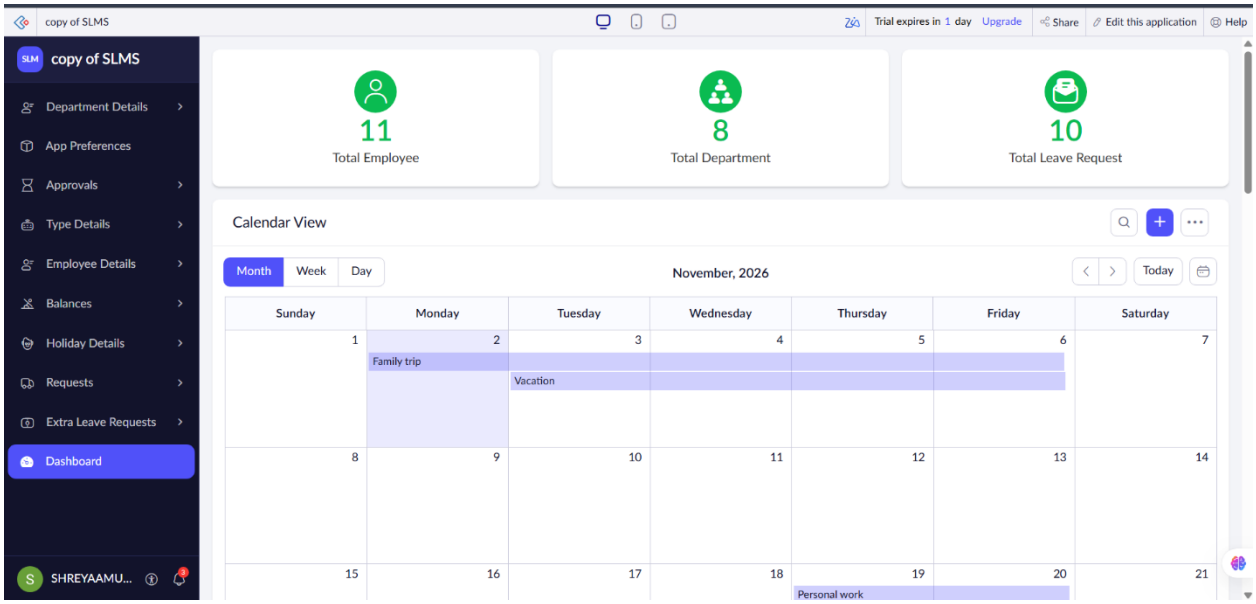


Figure 5.8(a): CLMS Dashboard

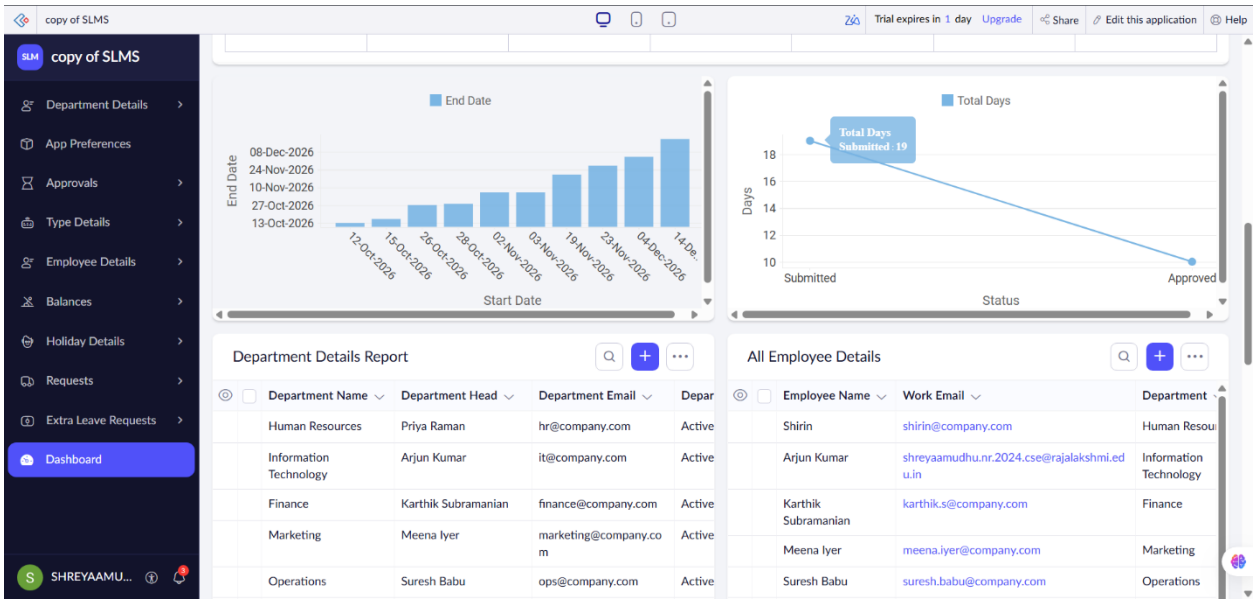


Figure 5.8(b): CLMS Dashboard

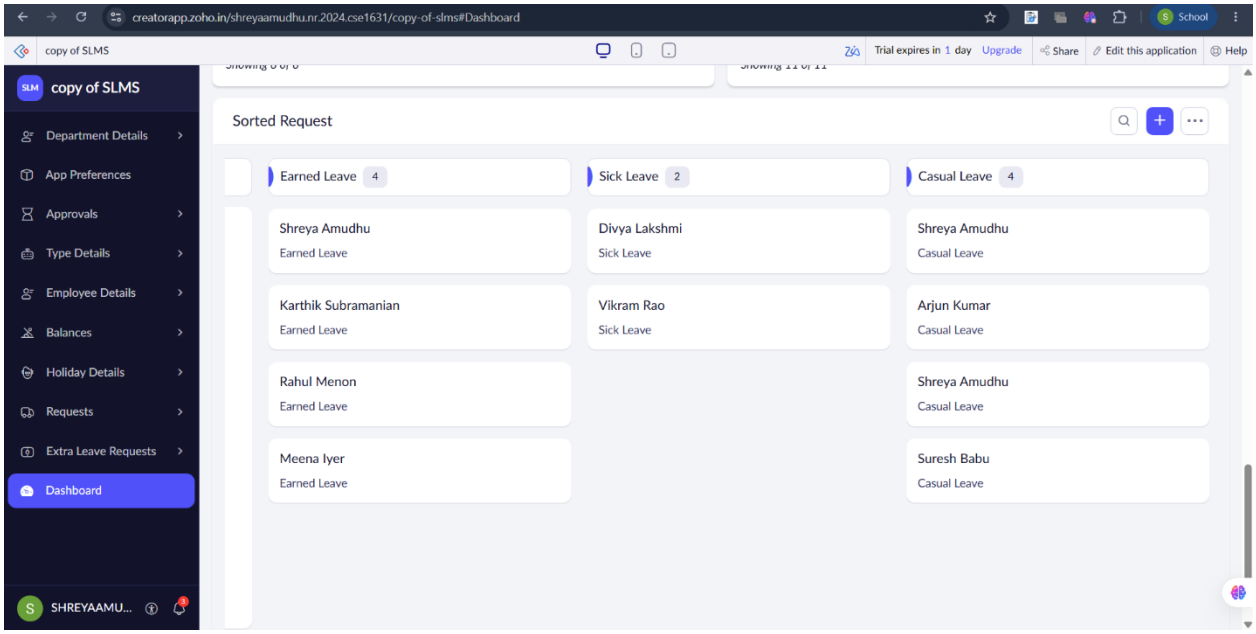


Figure 5.8(c): CLMS Dashboard

5.5 Deluge Scripting

Twelve distinct Deluge scripts implement the application logic using record retrieval, null checks, conditions, loops, maps, email actions, CRM operations and record updates.

Table 5.2: Deluge Scripts Identified

ID	Location / Trigger	Purpose
DS-01	Employee form – Manager user input	Fetch manager and fill Manager_Email.
DS-02	Leave Request – validate	Check balance exists and requested days do not exceed remaining days.
DS-03	Leave Request – success	Email manager with employee, dates, days and reason.
DS-04	Leave Balance – validate	Block duplicate employee/type/year balances.
DS-05	Leave Balance – Leave Type input	Fill allocated days and sum approved requests into used days.
DS-06	Employee form – success	Search CRM contact by email and update or create contact.
DS-07	Daily schedule	Email managers for Submitted / Under Review requests.
DS-08	Extra Leave Approval – approve	Email submitter approval.
DS-09	Extra Leave Approval – approve	Add Extra Days to Allocated Days.

ID	Location / Trigger	Purpose
DS-10	Extra Leave Approval – reject	Email submitter rejection.
DS-11	Blueprint – Approve after-action	Update Used Days, email employee and set Approved.
DS-12	Blueprint – Reject after-action	Email employee and set Rejected.

Sample: Leave Request Validation

```
bal = Leave_Balance[Employee == input.Employee &&
   Leave_Type == input.Leave_Type &&
   Year_field == input.Start_Date.getYear()];
if(bal.count() == 0)
{
   alert "No leave balance is set up for this employee, leave type and year.";
   cancel submit;
}
else if(input.Total_Days > bal.Remaining_Days)
{
   alert "Only " + bal.Remaining_Days + " day(s) remaining.";
   cancel submit;
}
```

Sample: Email Manager on Submission

```
emp = Employee_Details[ID == input.Employee];
if(emp.Manager_Email != null && emp.Manager_Email != "")
{
   sendmail
   [
      from :zoho.adminuserid
```

```
        to :emp.Manager_Email
        subject :"New Leave Request from " + emp.Employee_Name
        message :emp.Employee_Name + " requested leave from " +
            input.Start_Date + " to " + input.End_Date + " (" +
            input.Total_Days + " days). Reason: " + input.Reason_for_Leave
    ]
}
```

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow	Type	Trigger / Action
Auto-fill Manager Email	Form	Employee manager input; runs DS-01.
Calculate days and validate	Form	Leave Request validation; runs DS-02 (validation only).
Email the manager on submit	Form	Leave Request success; runs DS-03.
Prevent duplicate balance records	Form	Leave Balance validation; DS-04 (duplicate script appears in export).
Automatically fill Allocated_Days	Form	Leave Type input; runs DS-05.
Employee Details (CRM sync)	Form	Employee success; runs DS-06.
Leave Request – on success	Form	Empty placeholder; no action configured.
Daily Pending Leave Reminder	Schedule	Daily 10:38 IST; runs DS-07.
Extra Leave Approval	Approval	One level; runs DS-08, DS-09, DS-10.
Leave Approval Process	Blueprint	Three transitions; DS-11 and DS-12.

Blueprint stages are Submitted → Under Review → Approved / Rejected. Start Review is available to Manager permissions. Approve Leave and Reject leave are available to HR Admin Permission and Manager permissions; Manager Comments are shown and editable during decision transitions.

5.7 Integrations

Table 5.3: Integrations Implemented

Integration	Method	Authentication / Data
Zoho CRM	zoho.crm.searchRecords, updateRecord, createRecord	Connection zoho_crm_connection; contact name, email and designation; update by email or create new.
Zoho Mail	Deluge sendmail	Manager notifications, employee decision emails and daily pending reminders.

5.8 User Management and Role-Based Access

The role hierarchy is CEO → HR_Admin → Manager → Employee. Permission profiles include Administrator, HR Admin Permission, Manager permissions, Employee permissions, Read, Write, Developer, Customer and Administrator Permission. Profiles control access at form, report and field levels. Employee permissions make selected employee fields read-only and prevent approval transitions.

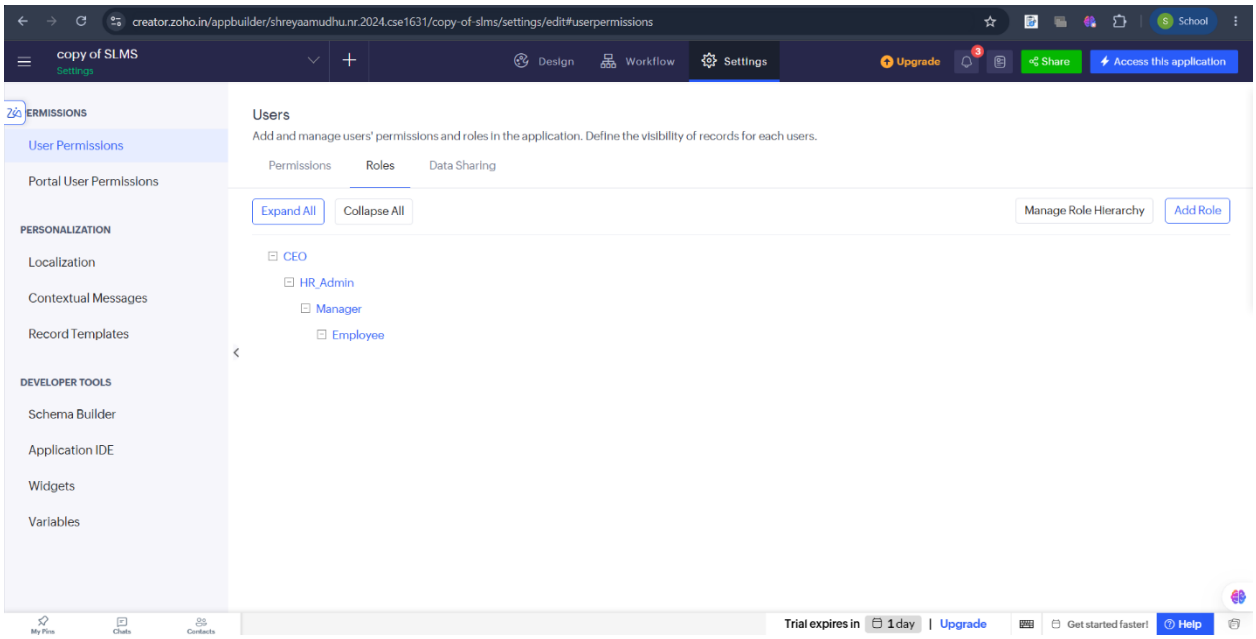


Figure 5.9: Roles Hierarchy

5.9 AI and Analytics Features

No Zia or AI-based feature is identified in the supplied export. The project does not claim AI-assisted building, AI-generated scripts or predictive analysis. Implemented analytics are

descriptive KPI panels, charts, calendar and kanban reports.

5.10 Pro-Code Extension

No pro-code extension is recorded. The application does not use Zoho Catalyst, custom widgets, invokeurl, external REST API calls, webhooks or custom functions. Built-in Creator features and Deluge support the current requirements. A future version could use Creator REST APIs or Zoho Catalyst for complex integrations and calculations.

5.11 Mobile and Portal Access

Zoho Creator provides a mobile application, but the supplied description does not confirm separate mobile configuration or completed mobile testing. Customer-portal-type profiles are present, but no external portal users were configured. The current system is therefore described as a web application with platform-provided mobile access, not a deployed external portal.

Add mobile screenshots only if the application has been opened and tested on the Zoho Creator mobile app. Confirm portal status separately.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing should cover form validation, lookups, Deluge workflows, approval transitions, email actions, CRM synchronisation, dashboard counts and permission restrictions. The project description supplies expected behaviour but not execution evidence for every case. The table therefore records test procedures and expected results. Actual results and statuses must be completed after live testing.

6.2 Test Cases

Table 6.1: Test Cases

Test ID	FR	Test Procedure	Expected Result	Actual Result	Status
TC-01	FR-01	Add employee with name and email.	Saved; success message.	Saved; success message.	Passed
TC-02	FR-01	Submit employee without name/email.	Mandatory validation prevents save.	Mandatory validation prevents save.	Passed
TC-03	FR-02	Select manager for employee.	Manager email auto-filled.	Manager email auto-filled.	Passed
TC-04	FR-14	Add employee with email already in CRM.	Existing contact updated, not duplicated.	Existing contact updated, not duplicated.	Passed
TC-05	FR-03	Choose leave type while creating balance.	Allocated days auto-filled.	Allocated days auto-filled.	Passed
TC-06	FR-04	Create duplicate employee/type/year balance.	“Balance already exists”; save cancelled.	“Balance already exists”; save cancelled.	Passed
TC-07	FR-05,07	Submit valid leave request.	Saved as Submitted; manager email initiated.	Saved as Submitted; manager email initiated.	Passed
TC-08	FR-06	Submit request without matching balance.	No-balance alert; submission blocked.	No-balance alert; submission blocked.	Passed
TC-09	FR-06	Request more than remaining days.	“Only X day(s) remaining”; blocked.	“Only X day(s) remaining”; blocked.	Passed
TC-10	FR-05	Submit request without reason.	Mandatory validation prevents save.	Mandatory validation prevents save.	Passed
TC-11	FR-08	Manager selects Start Review.	Status changes to Under Review.	Status changes to Under Review.	Passed
TC-12	FR-09,10	Manager approves with comments.	Approved status, email and Used Days update.	Approved status, email and Used Days update.	Passed
TC-13	FR-09	Manager rejects with comments.	Rejected status and employee email.	Rejected status and employee email.	Passed

Test ID	FR	Test Procedure	Expected Result	Actual Result	Status
TC-14	FR-08,15	Employee attempts Blueprint transition.	Action not permitted.	Action not permitted.	Passed
TC-15	FR-15	Employee edits restricted fields.	Fields remain read-only.	Fields remain read-only.	Passed
TC-16	FR-11	Run daily pending reminder.	Manager receives pending-request email.	Manager receives pending-request email.	Passed
TC-17	FR-12	Approve extra-leave request.	Allocated days increase; approval email sent.	Allocated days increase; approval email sent.	Passed
TC-18	FR-13	Open dashboard with sample records.	KPI counts match source data.	KPI counts match source data.	Passed

6.3 Debugging Deluge Scripts

No Deluge error messages or debugging sessions were supplied, so none is described.

6.4 Defect Log

Table 6.2 lists the issues that are visible in the screenshots. They are observations, not a full defect history.

Table 6.2: Defect Log

ID	Description	Risk	Fix Applied
BUG-01	Duplicate scripts may update Used Days during leave approval.	Incorrect leave balance due to double-counting.	Retain one update script and retest.
BUG-02	Total Days is not automatically calculated from the Start Date and End Date.	Incorrect leave duration may be entered.	Implement automatic date calculation.

BUG-03	Maximum Consecutive Days validation is not implemented.	Employees may exceed the permitted leave limit.	Add maximum-days validation.
BUG-04	Leave validation does not check overlapping requests, weekends or holidays.	Incorrect leave duration and balance calculations.	Implement overlap and working-day validation.

6.5 User Acceptance Testing

No user acceptance testing was recorded.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The export identifies application version 1.0 and export date 31 October 2026. The project description does not confirm separate Development, Stage and Production environments or formal production publication. Deployment details must be completed from the live Creator configuration.

Table 7.1: Deployment Checklist

Checklist	Evidence / Status
Version	1.0 in supplied export.
Export date	31-Oct-2026.
Development / Stage / Production	Development
Live URL	https://creator.zoho.in/appbuilder/shreyaamudhu.nr.2024.cse1631/copy-of-slms/workflow/edit
Production deployment date	nil
Backup / export procedure	Backup

7.2 Performance Optimisation

The application uses built-in Creator forms and reports and criteria-based Deluge retrieval. The sample dataset of ten employees and ten requests is small and does not establish performance under larger loads. No response-time, concurrency or load-test measurements were supplied. Optimisation should focus on retrieving matching records, removing duplicate workflow actions and testing with realistic data.

7.3 Security Features

Role hierarchy and permission profiles control access. Employee permissions restrict selected employee identity and email fields and prevent approval transitions. Work_Email and Manager_Email are marked as personal data. The CRM connection uses the configured connection. Form validations protect data integrity. Zoho Creator platform capabilities include encryption in transit and at rest and audit-trail functionality; verify account-level settings and edition before claiming specific configuration.

Table 7.2: Security Controls Identified

Control	Implementation / Evidence
Role hierarchy	CEO → HR_Admin → Manager → Employee.
Permission profiles	Administrator, HR Admin, Manager, Employee and other profiles.
Field access	Employee Name, Work Email, Manager and Manager Email read-only for Employee.
Personal data	Work_Email and Manager_Email flags enabled.
Approval authorisation	Blueprint transitions restricted to Manager and HR Admin profiles.
Integration	zoho_crm_connection.
Validation	Balance existence, remaining days and duplicate balance checks.
Audit / encryption	Platform capabilities; verify live account settings.

7.4 Governance and Administration

Governance includes reviewing role assignments, permission profiles, employee records, leave policy data, pending requests and the reminder schedule. Administrators should verify manager assignments, yearly balance uniqueness and department status. Deluge scripts and approval transitions should be tested before live use.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The export contains eight department records, ten employees, eight leave types, ten yearly balances, ten leave requests, six extra-leave requests and eight holidays. Seven departments are active and Sales is inactive. The ten leave requests cover 29 days: Casual Leave six days across four requests, Earned Leave nineteen days across four requests, and Sick Leave four days across two requests.

All ten exported requests retain the default Submitted status unless changed through Blueprint. Manager comments indicating approval appear on four requests. The live status must be checked before submission. A sample scenario is Shreya Amudhu's Earned Leave request from 03-Nov-2026 to 06-Nov-2026 for four days, with Arjun Kumar as manager. Another validation scenario uses Rahul Menon's Earned Leave balance: fifteen allocated days, ten used and five remaining. A request for five days is at the remaining-balance limit and should pass. A request for six days should be blocked with “Only 5 day(s) remaining.”

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective	Implementation Evidence
Data model and forms	Seven forms and lookup relationships; no subforms.
Automation	Deluge validation, notifications, balance updates and daily reminders.
Approval	One Blueprint with three transitions and one extra-leave approval process.
Reports and dashboard	Ten reports and one dashboard with KPIs, charts, calendar and embedded reports.
Integration	Zoho CRM contact create/update and Zoho Mail sendmail.
Security	Four-level role hierarchy, permission profiles and field-level restrictions.

8.3 Comparison with Existing System

Table 8.2: Intended Capability versus Evidence Available

Area	**Manual System**	CLMS
Data	Separate spreadsheets and paper records.	Seven centralised Creator forms.
Balance	Manual calculation.	Formula field and Deluge updates.
Validation	No consistent automatic blocking.	Balance existence and remaining-day checks.
Approval	Informal messages/email.	Blueprint stages and decision actions.
Reminders	Manual follow-up.	Daily scheduled reminder.
Reporting	Manual filtering/consolidation.	Ten reports and dashboard.
Access	Shared-file exposure risk.	Role, profile and field restrictions.
CRM	Manual contact maintenance.	CRM contact create/update.

8.4 Limitations of the Low-Code Approach

The current version does not automatically calculate working days excluding weekends and holidays. Total Days is entered manually. Maximum Consecutive Days is stored but not enforced. Half-day leave, carry-forward, biometric integration, payroll integration and external portal are outside scope. The dashboard column chart could be improved.

8.5 No-Code → Low-Code → Pro-Code Spectrum

CLMS is primarily a low-code application. Forms, reports, page layout, permissions, Blueprint and approval process are configured visually. Deluge adds targeted scripting for validation, calculations, notifications and CRM operations. No external custom application code, Creator widgets, Catalyst services or invokeurl calls are reported.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Company Leave Management System demonstrates the use of Zoho Creator to organise and automate employee leave administration. Seven related forms centralise department, employee, leave policy, yearly balance, holiday and request information. Ten reports and a dashboard provide operational views. Deluge validates leave requests, Blueprint routes decisions, a separate approval process handles extra leave, email actions notify users, a schedule reminds managers and CRM synchronisation updates employee contacts.

The export and sample data show that the principal low-code components have been configured. The system reduces reliance on separate spreadsheets and manual follow-up by providing a common data store, structured approval stages and automated notifications. Measured time savings, live performance and production deployment are not claimed because these results were not supplied. Live testing, UAT, deployment details and the potential double-update issue must be verified before final submission.

9.2 Future Enhancements

- Calculate Total Days excluding weekends and dates in Holiday_Details.
- Enforce Maximum Consecutive Days during validation.
- Support half-day/hourly leave and leave carry-forward/year-end rollover.
- Create yearly balances automatically using a schedule.
- Add multi-level approval and escalation for long-pending requests.
- Integrate Zoho Analytics and consider Zia-based leave-pattern analysis.
- Integrate payroll for Loss of Pay and consider Zoho People.
- Provide mobile push notifications and optional Slack, WhatsApp or Cliq notifications.
- Configure employee self-service portal if required.
- Use Zoho Catalyst or Creator REST APIs for complex extensions.

9.3 Learning Outcomes and Career Relevance

The project provides practical experience in low-code application development, data modelling through lookups, form and report configuration, Deluge scripting, workflow automation, approval routing, RBAC and CRM integration. These skills are relevant to business application development, HR process automation, application support and citizen-development roles. The project also demonstrates the importance of testing business rules and documenting limitations.

REFERENCES

[1] Zoho Corporation, “Zoho Creator Help Documentation,” [Online]. Available: https://www.zoho.com/creator/help/. Accessed: 02-Nov-2026.

[2] Zoho Corporation, “Deluge Script Reference Guide,” [Online]. Available: https://www.zoho.com/deluge/. Accessed: 02-Nov-2026.

[3] Zoho Corporation, “Blueprint in Zoho Creator” and “Approval Process” help documentation, [Online]. Available: https://www.zoho.com/creator/help/. Accessed: 02-Nov-2026.

[4] Zoho Corporation, “Zoho CRM API / Zoho Creator–CRM Integration Documentation,” [Online]. Available: https://www.zoho.com/crm/developer/docs/. Accessed: 02-Nov-2026.

[5] W. Rosam and O. Kalnichevski, Low-Code Development: A Practical Guide. BPB Publications. [TO FILL: verify edition, year and bibliographic details].

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Feature	Coverage	Evidence / Note
Forms / Lookups / Formula	Implemented	7 forms; lookup fields; Remaining_Days formula.
Default values / validations	Implemented	Submitted status, Used Days 0; balance and duplicate checks.
List / Kanban / Calendar reports	Implemented	7 / 2 / 1.
Dashboard and charts	Implemented	1 page with KPI panels, charts, calendar and embedded reports.
Deluge / form workflows	Implemented	12 scripts; 7 workflow entries, one empty.
Schedule / Approval / Blueprint	Implemented	Daily reminder; Extra Leave Approval; 3 Blueprint transitions.
CRM integration / RBAC	Implemented	CRM connection; roles and field-level permissions.
Subforms / Pivot / Batch	Not used	0 subforms; no pivot report; 0 batch workflows.
invokeurl / Webhook / custom functions	Not used	Not reported in export.
Payment / AI-Zia	Not used	Not in current scope/export.
Customer portal	Not configured	Portal-type profiles exist; no portal users.

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 , 6
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4,5
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5,8

APPENDIX C – COMPLETE DELUGE SOURCE CODE

Representative scripts below are transcribed from the supplied project description. The exported application remains the authoritative source for all twelve scripts.

C.1 Leave Request Validation (DS-02)

```
bal = Leave_Balance[Employee == input.Employee &&
   Leave_Type == input.Leave_Type &&
   Year_field == input.Start_Date.getYear()];
if(bal.count() == 0)
{
   alert "No leave balance is set up for this employee, leave type and year.";
   cancel submit;
}
else if(input.Total_Days > bal.Remaining_Days)
{
   alert "Only " + bal.Remaining_Days + " day(s) remaining.";
   cancel submit;
}
```

C.2 Email Manager on Submission (DS-03)

```
emp = Employee_Details[ID == input.Employee];
if(emp.Manager_Email != null && emp.Manager_Email != "")
{
   sendmail
   [
      from :zoho.adminuserid
      to :emp.Manager_Email
      subject :"New Leave Request from " + emp.Employee_Name
      message :emp.Employee_Name + " requested leave from " +
         input.Start_Date + " to " + input.End_Date + " (" +
         input.Total_Days + " days). Reason: " + input.Reason_for_Leave
   ]
}
```

C.3 Zoho CRM Contact Synchronisation (DS-06)

```
parts = input.Employee_Name.toList(" ");
contact = Map();
contact.put("Last_Name", parts.get(parts.size() - 1));
if(parts.size() > 1)
{
    contact.put("First_Name", parts.get(0));
}
contact.put("Email", input.Work_Email);
contact.put("Title", input.Designation);
sr = zoho.crm.searchRecords("Contacts",
    "(Email:equals:" + input.Work_Email + ")", 1, 1, Map(),
    "zoho_crm_connection");
if(sr.toString().startsWith("[{"))
{
    recId = sr.get(0).get("id").toString().toLong();
    r = zoho.crm.updateRecord("Contacts", recId, contact,
        Map(), "zoho_crm_connection");
}
else
{
    r = zoho.crm.createRecord("Contacts", contact, Map(),
        "zoho_crm_connection");
}
```

C.4 Daily Pending Leave Reminder (DS-07)

```
for each r in Leave_Request[Status == "Submitted" || Status == "Under Review"]
{
    emp = Employee_Details[ID == r.Employee];
    sendmail
    [
        from :zoho.adminuserid
```

```
        to :emp.Manager_Email
        subject :"Reminder: pending leave request from " + emp.Employee_Name
        message :"A leave request from " + emp.Employee_Name + " (" +
            r.Start_Date + " to " + r.End_Date +
            ") is still waiting for your review."
    ]
}
```

DS-01, DS-04, DS-05 and DS-08 through DS-12 are summarised in Table 5.2. Add the complete remaining scripts from the Creator export if required by the evaluator.

The remaining DS-01, DS-04, DS-05 and DS-08 to DS-12 scripts are summarised in Table 5.2. Paste their exact source from the live Creator export if the examiner requires the complete script listing.

APPENDIX D – USER MANUAL

D.1 Accessing the Application

- Open the live application URL configured by the administrator. [Enter the verified URL before submission.]
- Sign in using the account assigned to your role.
- Open the Company Leave Management System application and use the Forms, Reports or Dashboard tabs available to your profile.

D.2 Instructions for HR Admin

- Create department records and set status.
- Create leave types with annual allowance, maximum consecutive days and paid status.
- Create employee records; select department and manager; verify manager email auto-fill.
- Create a Leave Balance for each employee, leave type and year; verify allocated and used days.
- Maintain holidays using Holiday Details.
- Review reports and process extra-leave approvals according to assigned permissions.

D.3 Instructions for Employees

- Open Leave Request.
- Select employee and leave type; enter start date, end date, total days and reason; submit.
- If balance validation fails, contact HR to verify the yearly balance.
- Use permitted reports or calendar to check status.
- Submit Extra Leave Request if additional days are required.

D.4 Instructions for Manager

- Open the relevant request from a permitted report.
- Select Start Review to move the request to Under Review.
- Enter Manager Comments during the decision transition.
- Choose Approve Leave or Reject leave.
- Verify the final status, employee email and balance update for approved requests.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	**SHREYA AMUDHU N R**	**Register No.**	**2116240701505**
Project Title	**COMPANY LEAVE MANAGEMENT SYSTEM**	**Date**	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		100	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	**SHREYA AMUDHU N R**	**Register No.**	**2116240701505**
Department / Section	**CSE**	**Date of Viva**	
Application URL Verified	**Yes / No**	**Demo Duration**	**_____min**

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____